ASSIGNMENT SPECIFICATION & TECHNICAL DOCUMENTATION

K-EXPLORE — Campus Event Management Portal

A responsive event discovery, interactive filtering, and seat reservation platform built with semantic HTML5, modern CSS3, Tailwind CSS, and Vanilla JavaScript.

PLATFORM STATUS	CANDIDATE / USER	REPOSITORY	LIVE URL
PRODUCTION DEPLOYED	Kitonah	github.com/Kitonah/k-explore-events	kitonah.github.io/k-explore-events

1. Executive Summary & Problem Statement

The objective of the *K Explore – Web Designing* assignment is to engineer a modern, visually engaging, and fully responsive event management website. The platform enables university students and attendees to explore upcoming flagship events, inspect detailed schedules and agendas, and register via a verified booking form. The architecture emphasizes intuitive navigation, high legibility, accessible modal dialogs, and persistent local data storage.

2. Technical Architecture & Design Implementation

The project follows an organized, single-directory client architecture with zero package-manager build overhead, guaranteeing rapid static loading and cross-browser reproducibility.

```
k-explore-events/ ├── index.html # Semantic HTML5 structure, WAI-ARIA modal dialog, ticker tape ├──
```

3. Assignment Compliance & Evaluation Matrix

REQUIREMENT	CLASSIFICATION	ENGINEERING IMPLEMENTATION DETAILS
Hero Section	MANDATORY	Introduces campus activities with clear branding, action buttons, and active status indicators.
Flagship Event Card	MANDATORY	Dedicated spotlight section featuring the National Hackathon with real-time countdown metrics.
Event Catalog & Grid	MANDATORY	Responsive multi-card grid (1 col mobile, 2 cols tablet, 4 cols desktop) displaying dates, venues, categories, images, and descriptions.
Search & Filtering	MANDATORY	Instant multi-parameter search across titles, venues, and descriptions paired with category filter pills and an empty-state reset view.
Registration Modal	MANDATORY	Two-column responsive dialog displaying full agendas alongside a verified registration form that fits standard screens without cutoff.
Form Validation	MANDATORY	Client-side JavaScript validation for attendee name (≥ 3 chars), RFC 5322 email regex, and strict 10-digit phone number. Automatically links selected event.
Confirmation Feedback	MANDATORY	Dynamic animated toast notification system that issues a unique booking reference and updates card status.

REQUIREMENT	CLASSIFICATION	ENGINEERING IMPLEMENTATION DETAILS
Dark / Light Mode	BONUS FEATURE	System-synced color mode switcher with a pre-render head script to eliminate Flash of Unstyled Content (FOUC).
Live Countdown Clock	BONUS FEATURE	Synchronous countdown timer calculating Days, Hours, Minutes, and Seconds every 1,000 ms with a zero-floor clamp.
LocalStorage Storage	BONUS FEATURE	Browser-persisted booking registry allowing users to view, manage, and revoke active passes via a dedicated drawer.

4. Quality Assurance & Edge-Case Defenses

- **Double-Booking Prevention:** Checks existing storage records before writing to ensure the same email address cannot register twice for the same event.
- **Input Sanitization:** Automatic trimming of whitespace across names, emails, and phone inputs to prevent whitespace validation failures.
- **Modal Scroll Lock & Escape Handlers:** Body scroll locking prevents background jitter while modals are active, with full keyboard (Escape) dismissal.